

# Effective Independence: Quantifying the Reliability Ceiling of Same-Base-Model LLM Ensembles

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**Abstract**—Debate, self-consistency, and LLM-as-judge pipelines assume ensemble diversity they do not actually have when every voter is sampled from the same base model. We apply the classical effective-sample-size statistic  $N_{\text{eff}}$  to this setting and measure it across four diversity axes commonly used to build LLM ensembles (prompt, temperature, model size, and model family) on three reasoning and factuality benchmarks: GPQA-Diamond, MATH levels 4-5, and SimpleQA. Measured with the usual correlation coefficient, even  $N = 3$  ensembles behave like barely more than a single independent voter ( $N_{\text{eff}} = 1.09$  for prompt diversity, 1.51 for model-family diversity, 1.49 for all 12 configurations together). We then show that this coefficient flatters such ensembles, because it is bounded above by a function of the two members’ error rates: 11 of our 12 within-axis pairs sit exactly at that bound. Looking at the error sets directly explains why. In 53 of 66 configuration pairs, one member’s errors are a strict *subset* of the other’s, so the weaker member recovers nothing the stronger one missed and a majority vote has no material to work with. That nesting is total for pairs drawn from a single base model (28 of 28) and falls to about half for pairs spanning different labs (11 of 21), against a permutation baseline of roughly 15% ( $p < 0.0005$ ). The consequence is measurable at the level practitioners care about: majority voting fails to beat the single best member of its own ensemble on every axis we tested, and trails it by 12.5 points across all 12 configurations at twelve times the API cost. We recommend reporting the fraction of ensemble member pairs whose errors are nested alongside any correlation-based diversity claim, since it is a direct count that cannot be inflated by adding a weaker member.

**Index Terms**—large language models, ensemble diversity, self-consistency, LLM-as-judge, multi-agent debate, effective sample size, reliability

## I. INTRODUCTION

### A. Background and Motivation

Self-consistency decoding, multi-agent debate, and LLM-as-judge ensembles all treat each sampled response as an independent vote. In practice, many of these pipelines draw every sample from the same underlying base model, varying only a shallow surface property such as the system prompt or the sampling temperature. If those samples share correlated failure modes, the effective diversity of the ensemble is far smaller than its nominal size  $N$  suggests. Majority voting, confidence aggregation, and judge panels then inherit a reliability ceiling that is invisible to the people building them.

### B. Challenges in Quantifying Ensemble Diversity

Diversity is easy to induce along superficial axes (a different persona in the system prompt, a higher decoding temperature) and hard to induce along axes that plausibly decorrelate errors (a genuinely different training run, a different base model family). Prior evaluation practice rarely distinguishes between these axes’ contribution to ensemble reliability, and error correlation across ensemble members is rarely reported at all, leaving practitioners without a way to tell whether adding a 4th or 8th sample from the same model is buying real reliability or merely restating the 1st sample with different words.

### C. Role of $N_{\text{eff}}$ in This Work

We adapt the effective-sample-size framework used for correlated estimators to LLM ensembles. The quantity is Kish’s design effect [7], standard in survey sampling for discounting a nominal sample size by the correlation among its units: given  $N$  ensemble members with mean pairwise error correlation  $\bar{\rho}$ ,

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\bar{\rho}}. \quad (1)$$

$N_{\text{eff}}$  answers a concrete question: how many *independent* voters does this ensemble behave like? It also lets us compare diversity axes on a common scale rather than only on raw accuracy. We compute  $N_{\text{eff}}$  for 12 configurations spanning 4 axes (3 configurations per axis: prompt-diverse, temperature-diverse, size-diverse, and family-diverse), with bootstrap confidence intervals over questions, and report where each axis’s  $N_{\text{eff}}$  saturates as  $N$  grows.

### D. Contributions

- We decompose measured ensemble independence by *construction axis*, varying one practitioner-controllable choice at a time rather than auditing diversity in aggregate.
- We show the statistic normally used is biased in a specific, correctable direction:  $\phi$  is bounded above by a function of the members’ error rates (Eq. 2), so members differing in accuracy are credited with independence they do not have. 11 of our 12 within-axis pairs sit exactly at that bound.

- We identify the structure responsible: errors are largely *nested* rather than overlapping. In 53 of 66 pairs one member’s errors are a strict subset of the other’s, including all 28 same-base-model pairs, against a permutation baseline of roughly 15% ( $p < 0.0005$ ). A nested pair has zero complementary errors, so no aggregation rule over it can recover a question.
- We show the nesting rate falls monotonically as members become genuinely more different (100%, 82%, 52% for pairs sharing a model, sharing a lab, and spanning labs), which gives the practitioner guidance a correlation ranking only gestures at.
- We connect the structure to the outcome it predicts. Majority voting never beats the best member of its own ensemble on any axis, trailing it by up to 25 points, and the number of questions a better aggregation rule could have recovered is between 0 and 2 per ensemble.
- We compare five allocation strategies at equal API cost under all three measures, and show that the ranking a correlation-based  $N_{\text{eff}}$  would suggest does not survive the correction (a 38% apparent advantage for cross-family ensembling becomes about 1%). We release the harness, the graded error matrix, and a decision log recording every deviation from the frozen specification.

## II. RELATED WORK

Self-consistency decoding [1] samples multiple chain-of-thought [2] completions from a single model and takes a majority vote, implicitly treating the samples as independent draws. Multi-agent debate frameworks [3] extend this to multiple model instances arguing over several rounds, again typically instantiated with a single underlying model. LLM-as-judge evaluation [4] raises the same concern from the evaluation side: a judge panel built from one base model may share the judge’s own biases across every panelist. A parallel line builds explicit LLM ensembles for accuracy: Agent Forest [5] scales by sampling and voting over instances of one model, and Mixture-of-Agents [6] layers multiple LLMs. Neither measures the pairwise error correlation its aggregation step relies on, so a reported gain from more samples is not distinguishable, from the published results alone, from the diminishing-returns pattern we measure in Sec. IV.

Classical ensemble theory holds that accuracy gains are governed by the trade-off between member accuracy and pairwise error correlation: the ambiguity decomposition [8] formalized it, and Dietterich [9] and Kuncheva and Whitaker [10] survey the constructions that induce decorrelated errors and the diversity measures used to quantify them. We apply the same design-effect statistic to a setting that theory did not anticipate: API-only, black-box ensembles whose members cannot be inspected, retrained or explicitly decorrelated, only reconfigured along the surface axes an API caller is given.

Two concurrent 2026 studies confirm the independence assumption is unreliable, from different angles. Kim [11] audits five diversity metrics across 31,900 ensemble subsets on MMLU-Pro and finds that apparent diversity gains are largely

capability entanglement rather than genuine error decorrelation. Once capability is controlled for, simple majority voting beats the best individual model in under 10% of size-3 subsets. Ding [12] audits cross-model agreement as a confidence signal at large scale (265,000 samples on GPQA-Diamond and AIME) and reports only weak correlation (0.20–0.59) between agreement and correctness, concluding that self-consistency is a conditional proxy rather than a standalone confidence score. Both results corroborate our starting premise, that ensemble diversity is routinely overestimated, but neither decomposes *why*: they do not attribute the correlation they measure to specific, practitioner-controllable construction choices. Our contribution is narrower: we hold  $N_{\text{eff}}$  fixed and vary one structural axis at a time to localize *where* an ensemble builder’s choices buy real independence. We also sharpen Kim’s [11] conclusion. That work attributes apparent diversity gains to capability entanglement, established by controlling for capability across ensemble subsets. Sec. IV-C identifies a mechanism operating even within a fixed subset: the correlation coefficient is itself bounded by the members’ error rates, so two members of unequal accuracy *cannot* record high correlation however similarly they fail. The entanglement is therefore a property of the measurement, not only of the sampling, and it is correctable.

## III. METHODOLOGY

### A. Datasets

We evaluate on three public benchmarks spanning multiple-choice and free-response formats: GPQA-Diamond [13], a graduate-level multiple-choice science benchmark; MATH [14] restricted to levels 4–5 (the hardest tier); and SimpleQA [15], a short-form factual QA benchmark. Free-tier API quotas (Sec. V) constrained the question set to  $N = 30$  total (8 GPQA-Diamond, 11 MATH levels 4–5, 11 SimpleQA), a deliberate reduction from an original target of 800 questions. See `logs/decisions.md` for the two intermediate reductions (800→150→30) and the quota measurements that drove each.

### B. Diversity Axes and Configurations

We define 12 configurations across 4 diversity axes, 3 configurations per axis, per the frozen specification in `config/project_spec.yaml`:

- **Prompt-diverse**: one fixed model at temperature 0.7, with three system-prompt personas (“careful analyst”, “confident expert”, “skeptical reviewer”).
- **Temperature-diverse**: the same fixed model and a neutral prompt, at temperatures  $\{0.3, 0.7, 1.0\}$ .
- **Size-diverse**: one model family, 3 sizes (small, medium, flagship), neutral prompt, temperature 0.7.
- **Family-diverse**: 3 different model providers/labs, neutral prompt, temperature 0.7.

Each (question, configuration) pair receives exactly one sample, since the goal is to measure cross-configuration error correlation rather than within-configuration sampling variance.

One consequence is worth stating plainly: for the temperature axis, a single draw per configuration cannot separate temperature-induced diversity from ordinary sampling noise at that temperature, so that axis measures the diversity a practitioner actually gets from one sample per setting rather than the full sampling distribution.

### C. Grading

Each dataset is graded by its own deterministic procedure. GPQA-Diamond uses exact match on the extracted answer letter. MATH uses symbolic equivalence via `math_verify` on the extracted `\boxed{}` expression, so that algebraically equal forms are not counted as disagreements. SimpleQA uses normalized string containment against the reference answer, with an LLM judge as a fallback only when containment fails.

That fallback deserves an explicit caveat, because the judge is `gemini-3.5-flash`, which is also the base model behind 8 of the 12 configurations. A model therefore participates in grading responses that include its own. We mitigate this three ways. The judge is a fallback rather than the primary path, every invocation is written to an audit log released with the paper, and, as a robustness check, we recompute the paper’s main result with SimpleQA excluded entirely (Sec. IV-D). We would not use a self-judging setup by choice; it is a consequence of the free-tier constraint that no second model with sufficient daily quota was available.

### D. Effective Independence Metric

For any subset of configurations we compute the pairwise error-correlation matrix and  $N_{\text{eff}}$  via Eq. 1, with 95% confidence intervals from 1,000 bootstrap resamples over questions. We additionally report the fraction of questions on which  $\geq 80\%$  of an ensemble agrees and is wrong. This confident-correlated-error rate is the failure mode majority voting cannot detect or correct. Finally, we compare a fixed set of five allocation strategies at a matched budget of  $k = 3$  ensemble members (Sec. IV-F), which re-uses the cached results already generated for the axis comparison and issues no additional API calls.

### E. Experimental Harness

The harness, the frozen specification, the model configuration, the graded error matrix, and the full decision log are released with this paper at <https://github.com/reinamercy/Effective-Independence>. File paths given below refer to that repository. The decision log in particular records every deviation from the original specification, in the order the deviations happened and with the reasoning at the time.

All generations are produced through a cached, rate-limited API client (one persisted response per unique model/prompt/temperature/seed tuple), so no (question, configuration) pair is ever re-queried. Every API call is logged with the provider-returned model version string, token counts, and cost. All generation uses the free-tier model set frozen in `config/models.yaml`. The prompt- and temperature-diverse axes both use `gemini-3.5-flash` [16], which is

also the medium tier of size-diverse; the small and flagship tiers of size-diverse are `gemini-3.5-flash-lite` and `gemini-3.7-flash` [16]. Family-diverse combines `gemini-3.5-flash` [16], `z-ai/glm-5.2` [17], and `nvidia/nemotron-3-ultra-550b-a55b` [18]. (The OpenAI slot originally used `openai/gpt-oss-20b` but was replaced with `z-ai/glm-5.2` mid-project after OpenRouter permanently de-listed the former from its free tier; see the decision log.) Dataset loading, configuration construction, cached generation, per-dataset grading and  $N_{\text{eff}}$  analysis are independently rerunnable modules connected only through on-disk artifacts, so any stage can be re-run in isolation.

## IV. RESULTS

### A. Quantitative Analysis

All results are computed on a fixed dataset of 249 graded (question, configuration) cells, 69.2% of the  $30 \times 12$  design, at 73.1% overall accuracy. The absent cells were not dropped for any property of their content: they are the cells a per-model daily API quota did not permit us to collect (Sec. V), so missingness is driven by which model a configuration uses, not by question difficulty or by the answers. Because coverage is still uneven, we report both what the uneven data supports and what survives where every configuration is graded. Table I reports  $N_{\text{eff}}$  per axis with bootstrap 95% confidence intervals. Prompt-diverse and temperature-diverse, the two axes that vary only a surface property of one fixed base model, show the highest mean pairwise error correlation ( $\bar{\rho} = 0.87$  and  $0.77$ ) and collapse to  $N_{\text{eff}} \approx 1.1$ – $1.2$  regardless of their nominal size  $N = 3$ . Size-diverse (one family, three parameter scales) is only marginally better ( $\bar{\rho} = 0.73$ ,  $N_{\text{eff}} = 1.22$ ), consistent with `config/models.yaml`’s documented caveat that its “flagship” tier is a newer Flash-generation model rather than a genuinely larger one, so this axis likely understates what true parameter-scale diversity would buy. Family-diverse scores best on this metric: mean error correlation drops to  $\bar{\rho} = 0.50$  and  $N_{\text{eff}}$  rises to  $1.51$ , still far from the nominal  $N = 3$ . We deliberately stop short of calling this a settled axis ranking. Under partial coverage the family-diverse advantage rests on its two thinnest configuration pairs (Sec. IV-B), and those same two pairs turn out to be constrained by a marginal bound that accounts for their low correlation on its own (Sec. IV-C). Extending this to the full 12-configuration ensemble,  $N_{\text{eff}}$  rises only from about  $1.0$  at  $N = 2$  to  $1.49$  at  $N = 12$ : adding nine further configurations buys roughly half of one additional effective voter.

We also measure the confident-correlated-error rate: the fraction of questions where  $\geq 80\%$  of a  $\geq 3$ -member ensemble is simultaneously wrong. Across all 12 configurations this occurs on 5 of 30 eligible questions (16.7%), and per axis is highest for prompt-diverse (5/26, 19.2%). A majority vote over any of these ensembles returns the wrong answer confidently on those questions; diversity in surface presentation does not protect against it. Per-configuration accuracy varies almost as much within an axis as across axes (size-diverse spans 70% to 89%), which matters because an axis can spread its members

TABLE I  
 $N_{\text{EFF}}$  PER AXIS UNDER  $\phi$  (BOOTSTRAP 95% CI) AND UNDER THE  
 MARGINAL-ROBUST  $\bar{Q}$  (249/360 CELLS).  $n_{\text{MIN}}$  IS THE FEWEST QUESTIONS  
 GRADED FOR BOTH MEMBERS OF ANY PAIR IN THAT AXIS. THE  $\phi$   
 COLUMNS ARE INFLATED BY UNEQUAL MEMBER ACCURACY (SEC. IV-C).

| Axis             | $\bar{\rho}$ | $N_{\text{eff}}(\phi)$ | 95% CI       | $\bar{Q}$ | $N_{\text{eff}}(\bar{Q})$ |
|------------------|--------------|------------------------|--------------|-----------|---------------------------|
| Prompt-diverse   | 0.87         | 1.09                   | [1.00, 1.29] | 1.00      | 1.00                      |
| Temperature-div. | 0.77         | 1.18                   | [1.00, 3.00] | 1.00      | 1.00                      |
| Size-diverse     | 0.73         | 1.22                   | [1.03, 1.64] | 1.00      | 1.00                      |
| Family-diverse   | 0.50         | 1.51                   | [1.06, 1.74] | 0.99      | 1.01                      |
| All 12 configs   | 0.64         | 1.49                   | [1.17, 1.90] | 0.95      | 1.04                      |

widely on accuracy and still have highly correlated errors:  $N_{\text{eff}}$  responds to *which* questions each member gets wrong, not how many.

### B. How Much Data Each Correlation Rests On

Each pairwise correlation is computed only over questions where *both* members are graded, and that overlap varies sharply. Seven of the twelve within-axis pairs rest on 8–9 shared questions, two of them saturating at exactly  $\rho = 1.00$ , which over nine questions is more plausibly a small-sample artifact than a literal claim that two configurations never disagree. The five well-sampled pairs ( $n \geq 26$ ) sit in a much narrower band,  $\rho = 0.68$ – $0.91$ . This matters for the axis ranking: family-diverse’s advantage is driven by its two *thinnest* pairs ( $n = 8$ ,  $\rho = 0.38$ ;  $n = 9$ ,  $\rho = 0.40$ ), while its only well-sampled pair ( $n = 27$ ) has  $\rho = 0.71$ , squarely among the well-sampled same-base-model pairs. Sec. IV-D therefore turns to a structural measure that does not depend on estimating a coefficient at all.

### C. Correlation Is Capped by Accuracy Differences

A second constraint limits what the correlation numbers can support, and it cuts deeper than sample size. The phi coefficient between two binary error vectors is bounded by their marginals. If configuration  $A$  is wrong on a fraction  $p$  of questions and  $B$  on a fraction  $q$ , with  $p \leq q$ , then even when  $A$ ’s errors are a strict subset of  $B$ ’s, which is the most concordant arrangement available,

$$\phi_{\text{max}} = \sqrt{\frac{p(1-q)}{q(1-p)}}, \quad (2)$$

which equals 1 only when  $p = q$ . The bound follows from maximising  $\phi = (n_{11}n_{00} - n_{10}n_{01})/\sqrt{pq(1-p)(1-q)}$  over the  $2 \times 2$  tables compatible with fixed marginals: the numerator is largest when the smaller error set is contained in the larger, which drives one off-diagonal cell to zero and leaves Eq. 2. Two configurations with different accuracy therefore *cannot* record a high correlation, however similarly they fail. A low  $\phi$  is evidence of genuine error decorrelation only when it sits appreciably below this ceiling.

Checking each observed correlation against its own ceiling changes how Table I should be read. Eleven of the twelve within-axis pairs sit exactly at their marginal-bound ceiling:

their errors are already as correlated as their differing accuracy permits. This includes both pairs responsible for family-diverse’s advantage, where one member is far more accurate than the other ( $p = 0.12$  against  $q = 0.50$ , and  $p = 0.11$  against  $q = 0.44$ ). At those error rates Eq. 2 caps  $\phi$  at 0.378 and 0.395, and those are precisely the observed values. The low correlation on this axis is thus explained by its members having unequal accuracy, not by their failing on complementary questions.

Exactly one of the twelve within-axis pairs falls clearly below its ceiling: the two open-weight models, at  $\phi = 0.71$  against a bound of 0.86. That pair is also the best-sampled on its axis ( $n = 27$ ), and among the within-axis pairs it is the only evidence of error decorrelation not attributable to an accuracy mismatch. (Sec. IV-D widens the view to all 66 pairs, where cross-lab pairs supply further such cases.) We therefore do not claim that cross-family ensembling decorrelates errors. The defensible reading is narrower: measured correlation is uniformly close to the maximum the data permits, and what separates the axes in Table I is mostly how unequal their members’ accuracies are.

This does not undermine the  $N_{\text{eff}}$  values themselves, which describe the actual correlation structure whatever produces it, but it does undermine any causal reading of the axis ranking. It is also a caution: an ensemble can post a lower  $\bar{\rho}$ , and so a higher  $N_{\text{eff}}$ , simply by including a weaker member.

### D. Errors Are Nested, Not Complementary

The bound in Eq. 2 is tight precisely when one member’s errors are a subset of the other’s, so the finding that most pairs sit at their ceiling is really a claim about the *structure* of the error sets. That structure can be counted directly, without estimating a coefficient at all, and doing so turns out to be both more robust at this sample size and more informative about what an ensemble can do.

Call a pair *nested* when one member’s error set contains the other’s, i.e. one of the two off-diagonal cells of the  $2 \times 2$  error table is empty. Pairs in which either member has no errors at all over the shared questions are excluded rather than counted as nested, since such a member is trivially contained in anything and would inflate the rate. Nesting is the case that matters operationally: if  $A$ ’s mistakes are a subset of  $B$ ’s, then  $B$  never answers a question correctly that  $A$  gets wrong, so no aggregation rule over  $\{A, B\}$  can recover anything  $A$  missed; Sec. IV-E tests that consequence directly against majority voting. The pair is ordered by capability rather than diversified by it. A pair with complementary errors on both sides, by contrast, gives a vote something to work with.

The result is stark. In 53 of 66 configuration pairs the errors are fully nested (bootstrap 95% CI over questions [62%, 98%]). Table II breaks this down by how genuinely different the two members are, and the rate falls monotonically: **all 28 pairs drawn from the same base model are nested** (Wilson 95% CI [88%, 100%]), against 82% for pairs sharing a lab but not a model ([59%, 94%]), and 52% for pairs spanning different labs ([32%, 72%]). A Cochran–Armitage test across the three

ordered groups confirms the trend rather than leaving it to the eye ( $z = -4.12$ ,  $p < 10^{-4}$ ). Fig. 1 plots the rates.

One scoping caveat belongs here, not only in Limitations. Eight of our twelve configurations run on `gemini-3.5-flash`, so the same-model row characterises one model observed under eight configurations rather than a property established across many base models. Whether it generalises is an open question our model set cannot answer.

Sparse tables could produce empty off-diagonal cells by luck, so this rate means nothing without a baseline. We therefore permute, holding each configuration’s error count and its set of graded questions fixed and shuffling only *which* of those questions it fails. That destroys any real shared failure mode while preserving the marginals and the coverage pattern exactly. Over 2,000 permutations the expected nesting rate is 15% (95% range [8%, 26%]) across all pairs and 16% (95% range [4%, 32%]) among same-model pairs. The observed 80% and 100% are roughly five times chance, and no permutation in either null reached the observed value ( $p < 0.0005$ ). The nesting is a property of the models, not of the sample size.

A second control addresses the grading circularity of Sec. III-C. Excluding SimpleQA entirely leaves 19 questions and 21 defined pairs, where nesting is 86% (18 of 21) and same-model pairs remain fully nested (3 of 3). The effect is not produced by the self-judged dataset, though that subset is small enough that we offer it only as a check on direction.

The first row is the paper’s sharpest result. Across every pair sharing a base model there is not one question the second configuration gets right that the first gets wrong. Changing persona or temperature reorders and rewords the output, and changes how often the model is wrong, but never contributes an independent success. That is a stronger and more checkable claim than  $\bar{\rho} = 0.87$ , and it does not rest on estimating a correlation from nine questions.

It also explains the earlier numbers. Nested pairs are exactly those pinned to the Eq. 2 ceiling, so the  $N_{\text{eff}}$  spread in Table I largely tracks how unequal the members’ accuracies are while the underlying dependence is close to total throughout. Recomputing with Yule’s  $Q$ , which is not marginal-bounded and reaches 1 whenever an off-diagonal cell is empty, makes this explicit: every axis falls to  $N_{\text{eff}} \approx 1.0$  (Table II), the 12-configuration ensemble to 1.04.

We are explicit about that substitution’s status. Eq. 1 is derived for a pairwise *correlation*, and  $Q$  sits on a different scale, so  $N_{\text{eff}}(Q)$  is a diagnostic rather than a drop-in estimator: it shows that the ordering and magnitudes implied by  $\phi$  do not survive a measure free of the marginal cap.  $Q$  also saturates at exactly 1 whenever an off-diagonal cell is empty, so with six or seven errors in some pairs a single flipped answer would move it. The nesting counts, which involve no formula, therefore carry the claim;  $N_{\text{eff}}(Q)$  expresses it on a familiar scale. Both point the same way, and further than  $\phi$  does.

#### E. Does Majority Voting Actually Help?

Everything above characterises the ensemble through a statistic or a structure. Neither is what a practitioner buys. The

TABLE II  
ERROR NESTING (249/360 CELLS). A PAIR IS NESTED WHEN ONE MEMBER’S ERRORS ARE A SUBSET OF THE OTHER’S, SO THE PAIR HAS NO COMPLEMENTARY ERRORS FOR A VOTE TO EXPLOIT.

| Pair type                 | Nested pairs | Rate |
|---------------------------|--------------|------|
| Same exact model          | 28 / 28      | 100% |
| Same lab, different model | 14 / 17      | 82%  |
| Different lab             | 11 / 21      | 52%  |
| All pairs                 | 53 / 66      | 80%  |

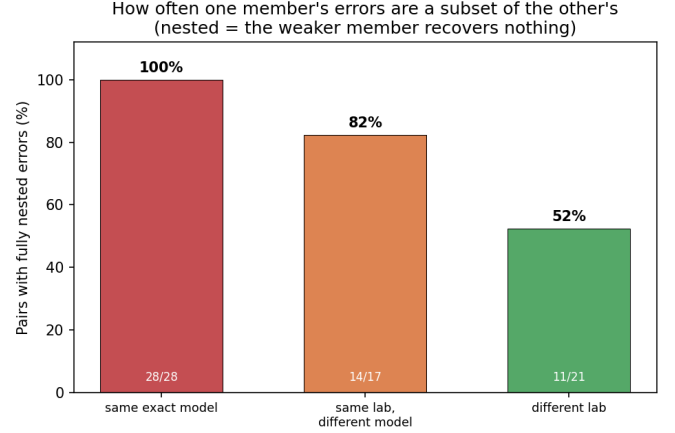


Fig. 1. How often one member’s errors are a strict subset of the other’s, by how genuinely different the two members are (249/360 cells).

question they face is whether aggregating  $k$  samples answers more questions correctly than using one model, so we measure that directly. It requires no additional generation: majority vote is computed from the same graded cells, with an even split counted as incorrect. To keep the panel constant across questions, each ensemble is scored only on questions where all of its members are graded.

Table III reports the result, and it is one-sided. **Majority voting does not beat the single best member of its own ensemble on any axis.** It loses by 3.8 points on prompt-diverse, 11.1 on temperature-diverse and 25.0 on family-diverse, and ties on size-diverse. Over all 12 configurations together it reaches 75.0% against 87.5% for the best member, a deficit of 12.5 points at twelve times the API cost.

This is the outcome the nesting structure predicts. When one member’s errors are a subset of another’s, the ensemble is ordered by capability, and a vote can only dilute the strongest member with weaker ones that contribute no question it missed. The recoverable column makes the size of the missed opportunity explicit: across all five ensembles, the number of questions where the majority was wrong while some member was right is 0 to 2. There is very little for a better aggregation rule to find, so the failure is not a deficiency of majority voting specifically. It is the absence of complementary errors to aggregate.

The obvious objection is that the best member is identifiable only in hindsight; a practitioner choosing in advance is more

fairly compared against the *mean* member, which voting matches or beats everywhere except family-diverse. The honest summary is narrower than “voting is useless”: voting recovers roughly the average member’s accuracy, never an accuracy no individual member already had.

TABLE III

MAJORITY VOTE AGAINST THE MEMBERS IT AGGREGATES (249/360 CELLS). “RECOV.” COUNTS QUESTIONS WHERE THE MAJORITY WAS WRONG BUT AT LEAST ONE MEMBER WAS RIGHT. A NON-POSITIVE VOTE—BEST COLUMN MEANS VOTING NEVER BEAT THE STRONGEST MEMBER.

| Ensemble            | $n_q$ | Vote | Best | Mean | Recov. |
|---------------------|-------|------|------|------|--------|
| Prompt-diverse      | 26    | 76.9 | 80.8 | 76.9 | 1      |
| Temperature-diverse | 9     | 77.8 | 88.9 | 81.5 | 1      |
| Size-diverse        | 9     | 88.9 | 88.9 | 81.5 | 0      |
| Family-diverse      | 8     | 62.5 | 87.5 | 62.5 | 2      |
| All 12 configs      | 8     | 75.0 | 87.5 | 71.9 | 1      |

#### F. Budget-Matched Allocation

The axis comparison above asks which kind of diversity decorrelates errors. The question a practitioner actually faces is narrower: given a fixed number of API calls per question, how should they be spent? We therefore compare five allocation strategies at a matched budget of  $k = 3$  calls. Every strategy issues exactly the same number of requests, so any difference in  $N_{\text{eff}}$  is attributable purely to how the budget was allocated rather than to how large it was.

We note explicitly that the frozen specification framed this comparison as one at matched total API *cost*. Every model in the final configuration is free-tier, so dollar cost cannot be the matching variable; we match on  $k$  instead, which is both the genuinely scarce resource in this project (per-model daily request quotas) and the quantity a practitioner controls. This substitution is recorded in `logs/decisions.md` rather than made silently.

Table IV reports each allocation under all three measures. Reporting only the  $\phi$  column here would contradict Sec. IV-C, and the disagreement between the columns is large enough that it changes the conclusion.

Read through  $\phi$ , cross-family looks decisively best:  $N_{\text{eff}} = 1.51$  against 1.09 for three prompts of one model, a 38% gain at identical cost. Almost all of that gap is the accuracy artifact. Under  $Q$  the same comparison is 1.01 against 1.00. We therefore do not claim a 38% reliability gain from crossing model families, and a practitioner should not budget for one.

What survives the correction is categorical, and visible in the last column. Every allocation built from a single base model is fully nested, 3 pairs of 3: no member ever answers a question its partners miss. The two allocations spanning more than one model each contain exactly one pair with complementary errors. That is the real distinction at a fixed budget, and it is a difference in kind, not a 38% improvement in a statistic.

This also corrects a reading  $\phi$  invites. Under  $\phi$  the mixed allocation (1.17) looks far worse than cross-family (1.51), sug-

gesting that spreading a budget across axes poorly substitutes for changing model family. Under  $Q$  the two are indistinguishable (1.01 each) and both contain one complementary pair. What matters is whether the allocation spans more than one base model at all, not how many *kinds* of diversity it mixes. With three effective ties and  $N = 30$ , we cannot rank the non-nested allocations against each other and do not attempt to.

TABLE IV

ALLOCATIONS AT A MATCHED BUDGET OF  $k = 3$  API CALLS PER QUESTION (249/360 CELLS). EVERY ROW COSTS THE SAME. THE  $\phi$  COLUMN OVERSTATES THE SPREAD BECAUSE IT IS CAPPED BY DIFFERENCES IN MEMBER ACCURACY (SEC. IV-C); THE  $Q$  AND NESTING COLUMNS ARE THE ONES TO READ.

| Allocation ( $k = 3$ )     | $N_{\text{eff}}(\phi)$ | $N_{\text{eff}}(Q)$ | Nested     |
|----------------------------|------------------------|---------------------|------------|
| Same model, 3 prompts      | 1.09                   | 1.00                | 3/3        |
| Same model, 3 temperatures | 1.18                   | 1.00                | 3/3        |
| Same family, 3 sizes       | 1.22                   | 1.00                | 3/3        |
| Mixed, one per axis        | 1.17                   | 1.01                | <b>2/3</b> |
| 3 different families       | 1.51                   | 1.01                | <b>2/3</b> |

#### V. LIMITATIONS

**Model set.** Budget constraints limited us to free-tier providers (Sec. III-E) rather than the closed-frontier set originally planned, after we found the real free-tier quota (20 requests/day/model) far below what public documentation reports. Family-diverse therefore mixes closed-weight (Gemini) and open-weight (GLM, Nemotron) models rather than three frontier flagships, and its members are the most unequal in accuracy of any axis, which interacts with the bound below. **Dataset size.** At  $N = 30$  questions ( $\sim 8$ –11 per dataset) some 95% intervals span nearly their whole range (temperature-diverse: [1.00, 3.00]). This is a power limitation, not an artifact of the method, so results should be read as a pilot. Intervals tighten as coverage grows (prompt-diverse moved from [1.00, 3.00] at 205/360 cells to [1.00, 1.29] at 249/360) but will not all become informative at  $N = 30$ . **Uneven coverage.** Missing cells concentrate in the eight configurations sharing one model’s quota, so graded counts range from 9 to 30 and most pairs rest on 8–9 shared questions. Cross-axis comparisons are therefore confounded with coverage. A complete-case analysis removes the confound but leaves only 8 of 30 questions with all twelve configurations graded, too few to rank four axes; we report the all-12 voting result on those 8 questions (Table III) and otherwise decline to rank. Table I is descriptive of this dataset, not a population ordering. **Both association measures are constrained.**  $\phi$  cannot exceed the Eq. 2 ceiling and 11 of 12 within-axis pairs sit exactly there, so differences in  $\bar{\rho}$  largely track how unequal member accuracies are; a design holding accuracy constant would separate the effects, but the free-tier set fixes which models exist at each tier.  $Q$  conversely saturates at 1 whenever an off-diagonal cell is empty, and with six or seven errors in some pairs one flipped answer would move it. The nesting counts are immune to both problems. **Nesting is pairwise.** Three configurations can be

pairwise nested in a chain with no single member dominating all others; our counts do not distinguish that case, and the operational claim we make is the pairwise one. **Single sample per cell.** We measure cross-configuration correlation, not within-configuration sampling variance, so for the temperature axis in particular one draw per setting cannot separate temperature-induced diversity from sampling noise. **A grading judge shares a base model with the configurations.** SimpleQA falls back to an LLM judge when string containment fails, and that judge is `gemini-3.5-flash`, which underlies 8 of the 12 configurations (Sec. III-C). Excluding SimpleQA leaves the nesting result intact (86% overall, same-model pairs still fully nested), so we do not believe it drives the finding, but an independent grader would be preferable and was unavailable within the quota.

## VI. CONCLUSION

On the data collected here, prompt- and temperature-diversity are the cheapest and most commonly used axes, and the two that buy the least real independence:  $N_{\text{eff}} \approx 1.1\text{--}1.2$  at nominal  $N = 3$ . Model-family diversity is the axis on which measured correlation falls furthest, but we do not claim the gain its  $\phi$ -based  $N_{\text{eff}}$  of 1.51 implies: that margin rests on the thinnest pairs (Sec. IV-B) and on pairs whose low correlation is already explained by unequal member accuracy (Sec. IV-C). Corrected, the comparison is 1.01 against 1.00.

The ranking is the least robust part of these results. The sturdiest finding is structural and needs no coefficient: in 53 of 66 pairs one member’s errors are a strict subset of the other’s, including every pair drawn from the same base model. Where errors are nested there is nothing to recover, which is why a nominal three or twelve votes behaves like roughly one. The prediction is borne out directly: majority voting never beat the best member of its own ensemble, and only 0 to 2 questions per ensemble were even available for a smarter aggregator to rescue.

For debate, self-consistency, and judge-panel pipeline design, that means scaling the number of same-base-model samples is a much weaker reliability lever than its nominal  $N$  suggests. At a fixed budget the distinction our data supports is categorical rather than a matter of degree: allocations drawn entirely from one base model are fully nested and can recover nothing, while allocations spanning more than one base model contain at least one pair with complementary errors. How many different *kinds* of diversity are mixed matters less than whether the base model changes at all.

Measuring that structure carries its own trap: because  $\phi$  is capped by the members’ error rates, an ensemble can report a flattering  $N_{\text{eff}}$  merely by including a weaker member. The remedy needs no extra API calls. Alongside any reported  $\bar{\rho}$  or  $N_{\text{eff}}$ , publish the fraction of member pairs whose errors are nested: a direct count, uninflatable by adding a weak member, answering what a builder needs to know.

These are pilot-scale results (Sec. V), consistent in direction with classical ensemble theory and with concurrent independent work [11], [12]. The magnitudes, and the axis ranking

especially, remain provisional until coverage is complete. The nesting result is the one we expect to survive, since further coverage can only add questions on which a member might differ, and 28 of 28 same-model pairs have produced none so far.

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